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(19) **United States**(12) **Patent Application Publication**
Luger(10) **Pub. No.: US 2025/0276231 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **UPPER BODY SKI MOTION EXERCISE MACHINE***A63B 21/16* (2006.01)*A63B 23/12* (2006.01)*A63B 71/06* (2006.01)(71) Applicant: **TORQUE FITNESS, LLC**, Coon Rapids, MN (US)(52) **U.S. Cl.**CPC *A63B 69/18* (2013.01); *A63B 21/153*(2013.01); *A63B 21/16* (2013.01); *A63B 23/12*(2013.01); *A63B 21/00069* (2013.01); *A63B**71/0619* (2013.01); *A63B 2210/00* (2013.01)(72) Inventor: **Nathan Luger**, Roseville, MN (US)(21) Appl. No.: **18/597,153**(22) Filed: **Mar. 6, 2024**

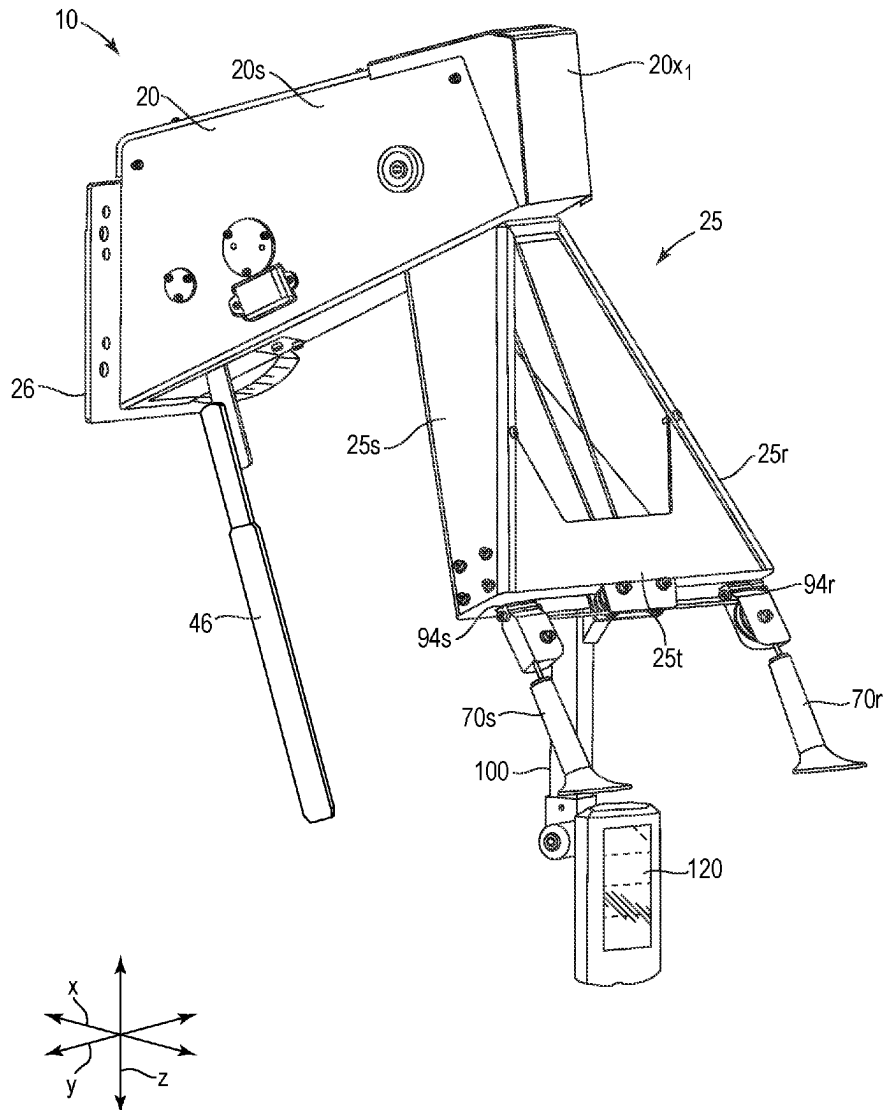
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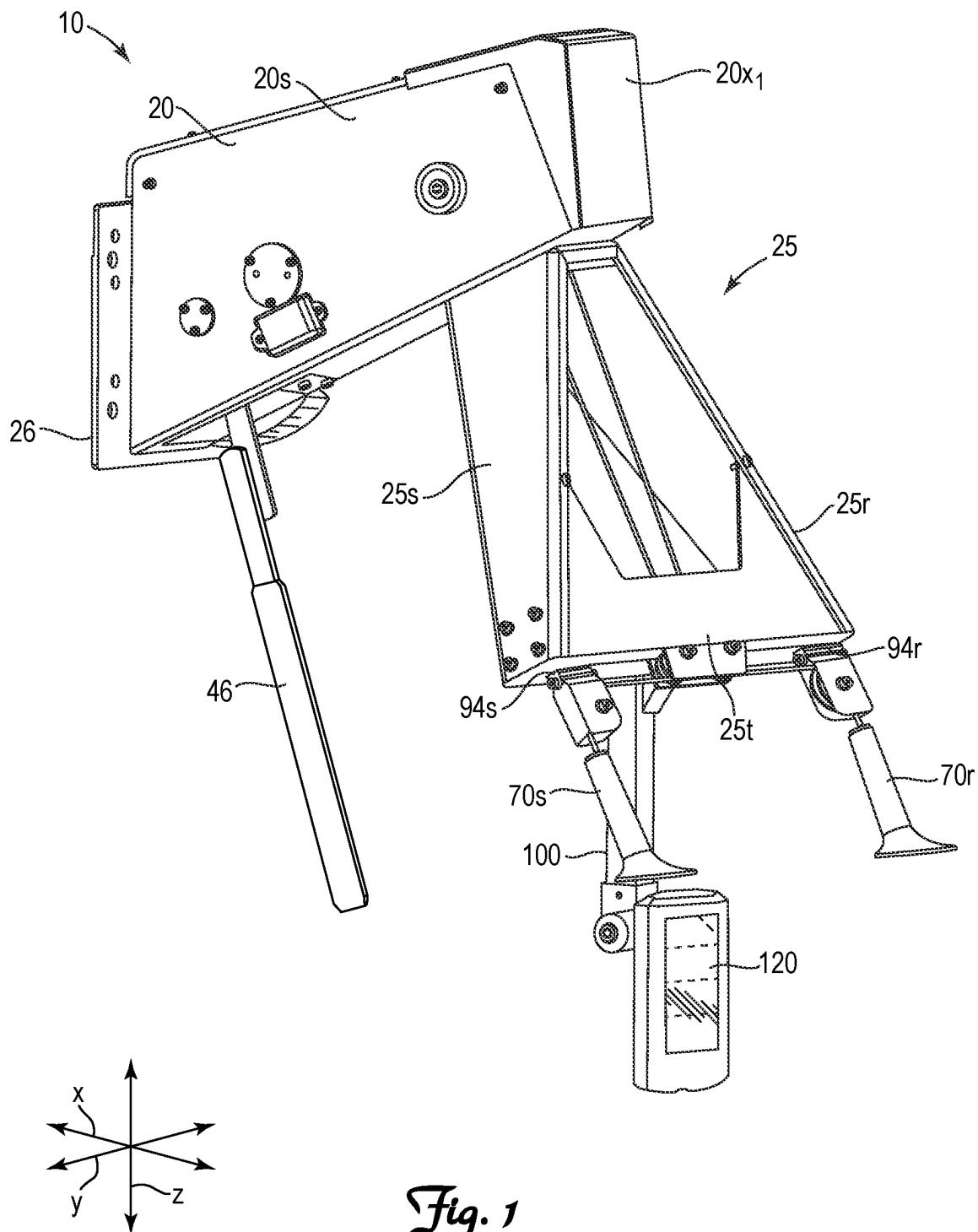
ABSTRACT**Related U.S. Application Data**

(60) Provisional application No. 63/560,300, filed on Mar. 1, 2024.

Publication Classification(51) **Int. Cl.***A63B 69/18* (2006.01)*A63B 21/00* (2006.01)

An upper body ski motion exercise machine that includes right and left spools, a single means for applying resistance to rotation of the spools, right and left side elastic lines each having one end anchored and the other end secured to the respective spool, right and left side inelastic lines each having one end secured to the respective spool and a grip attached to the other end, wherein the elastic and inelastic lines are configured and arranged for inverse winding and unwinding on the spool.





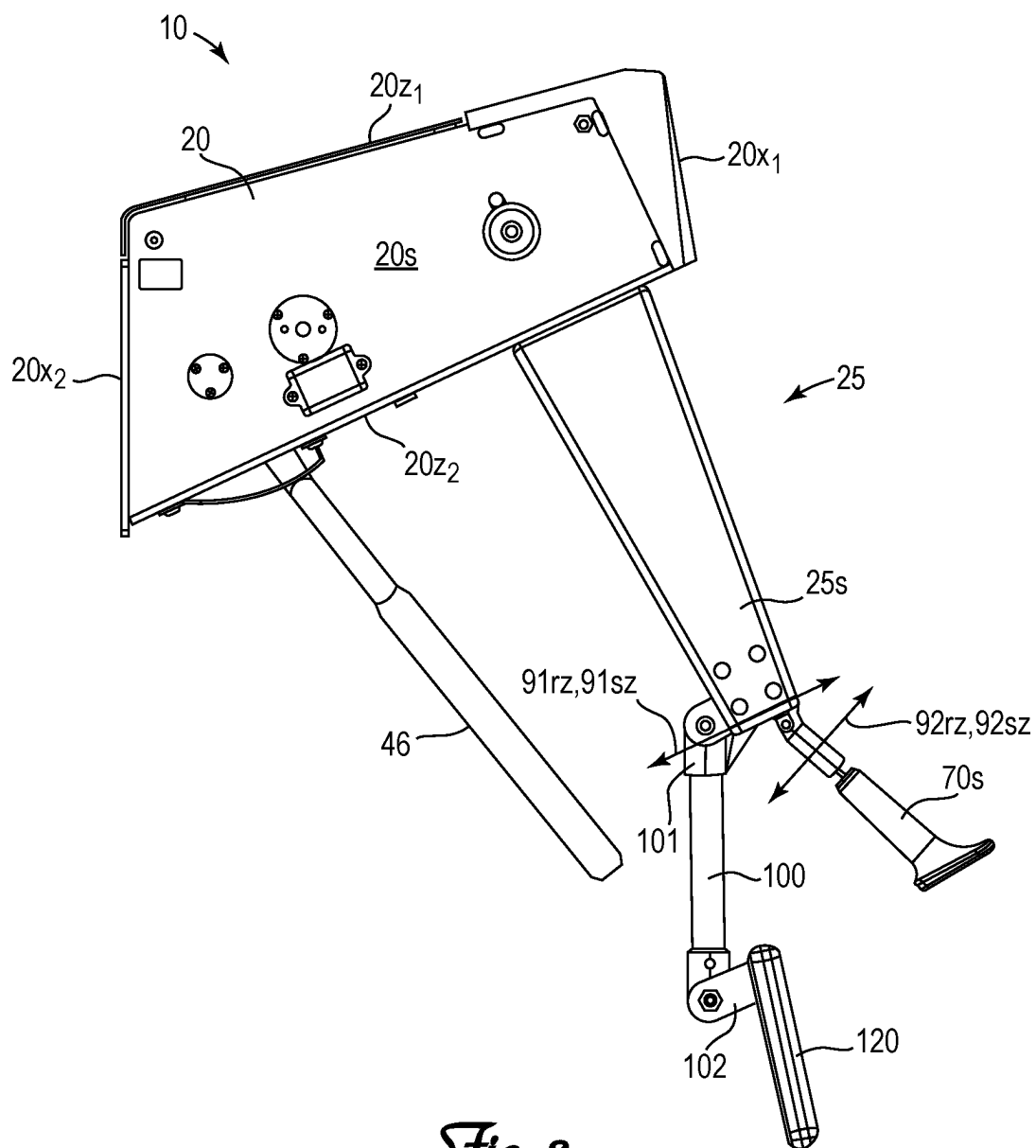


Fig. 2

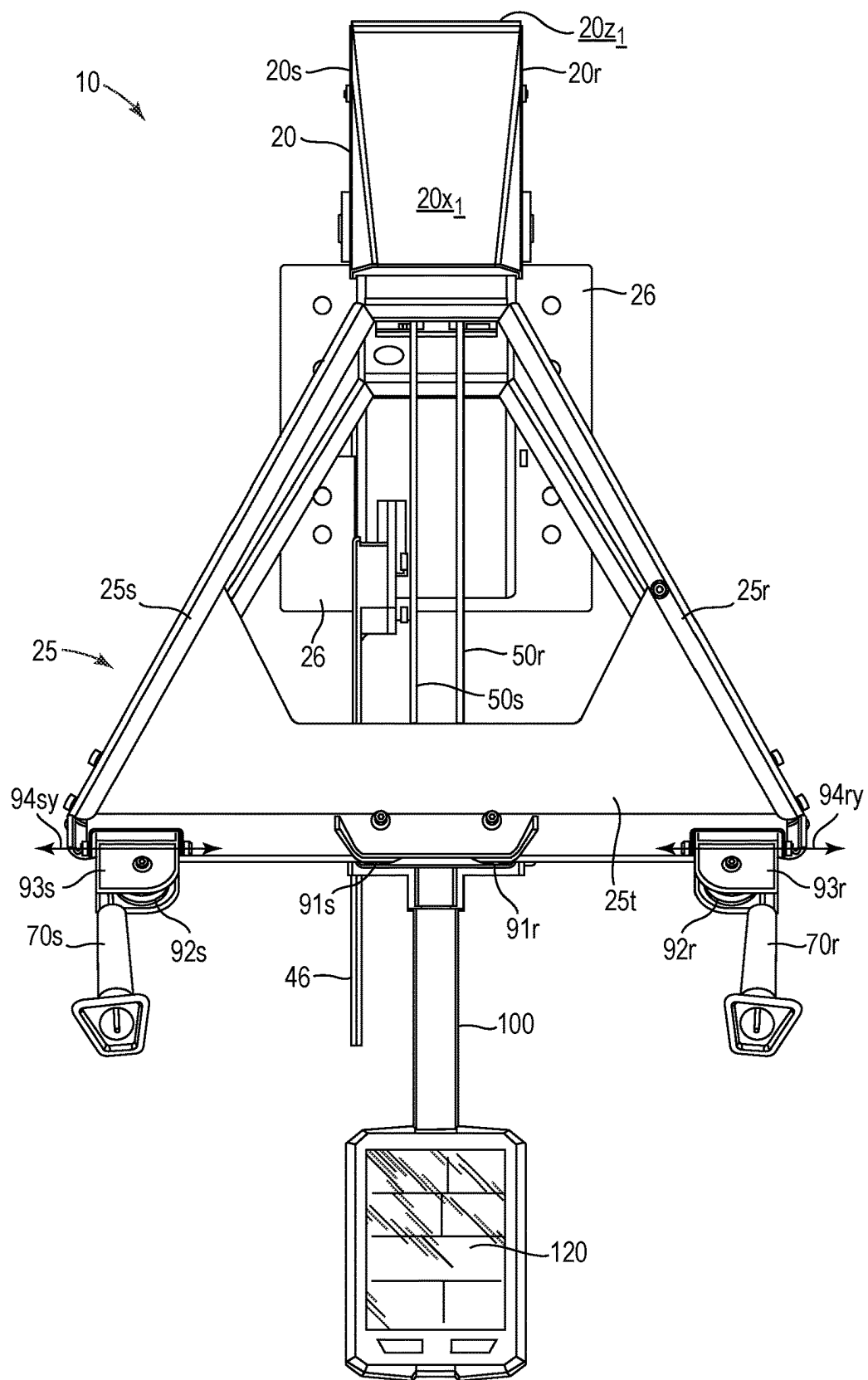


Fig. 3

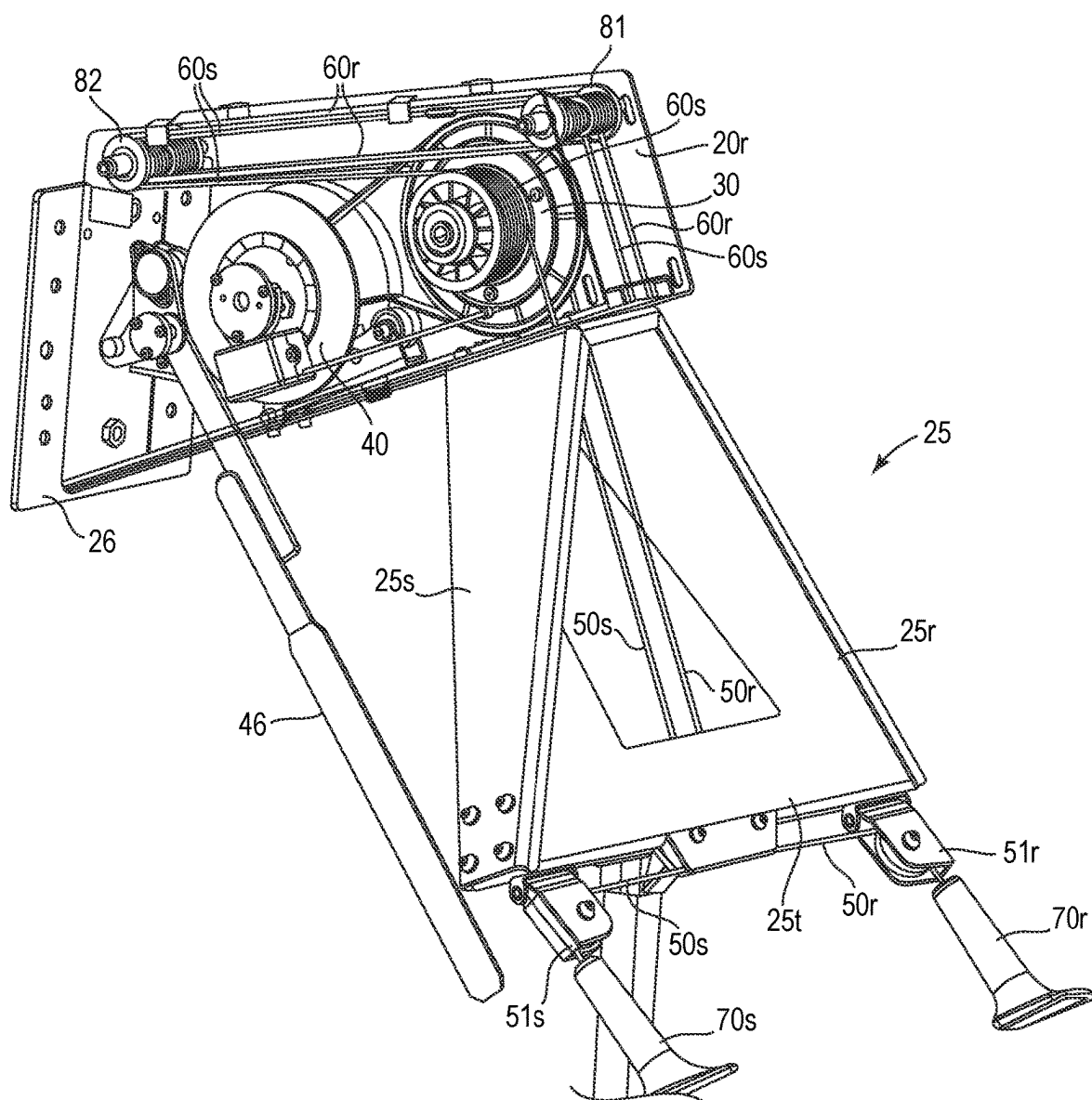


Fig. 4

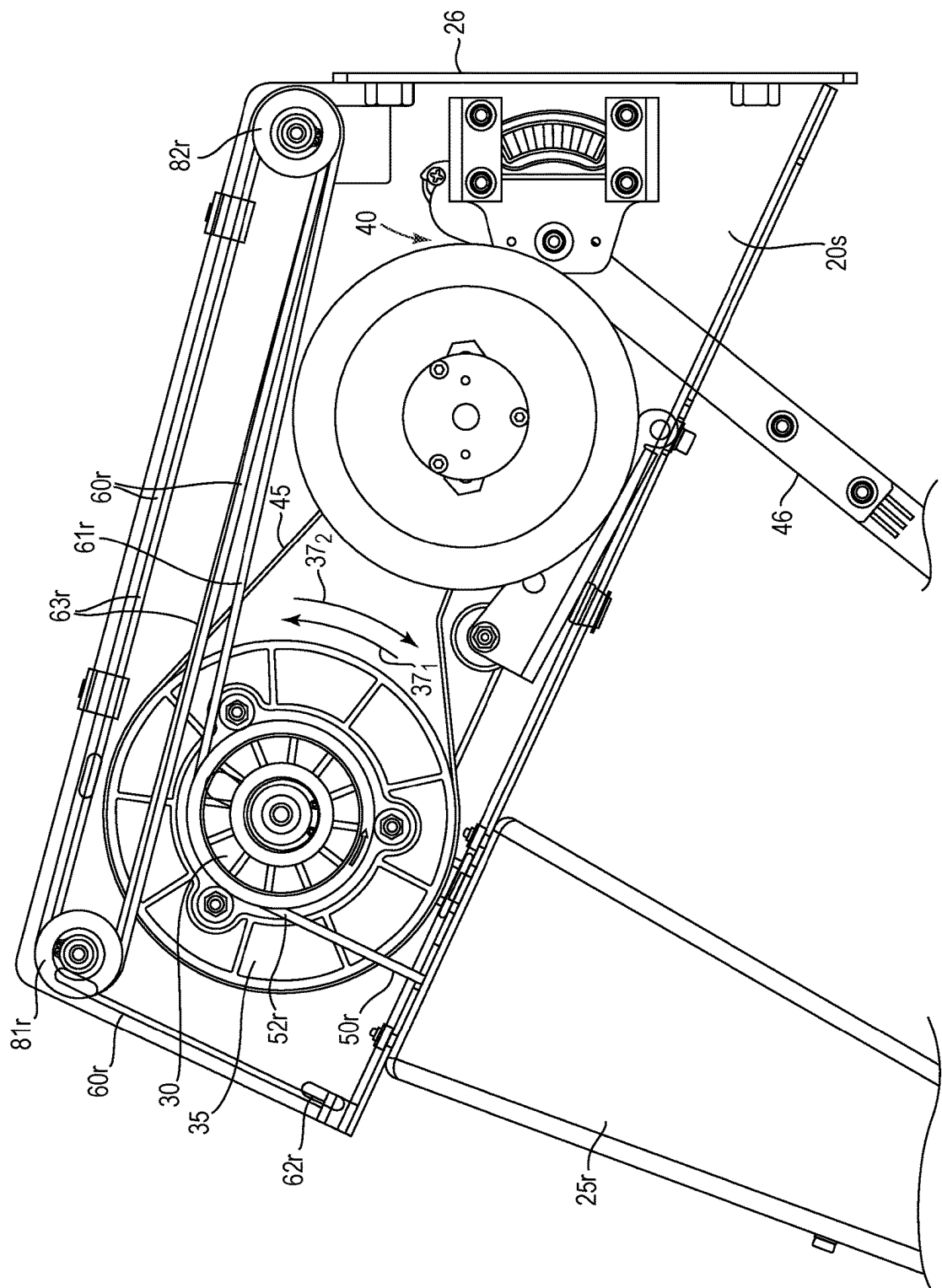
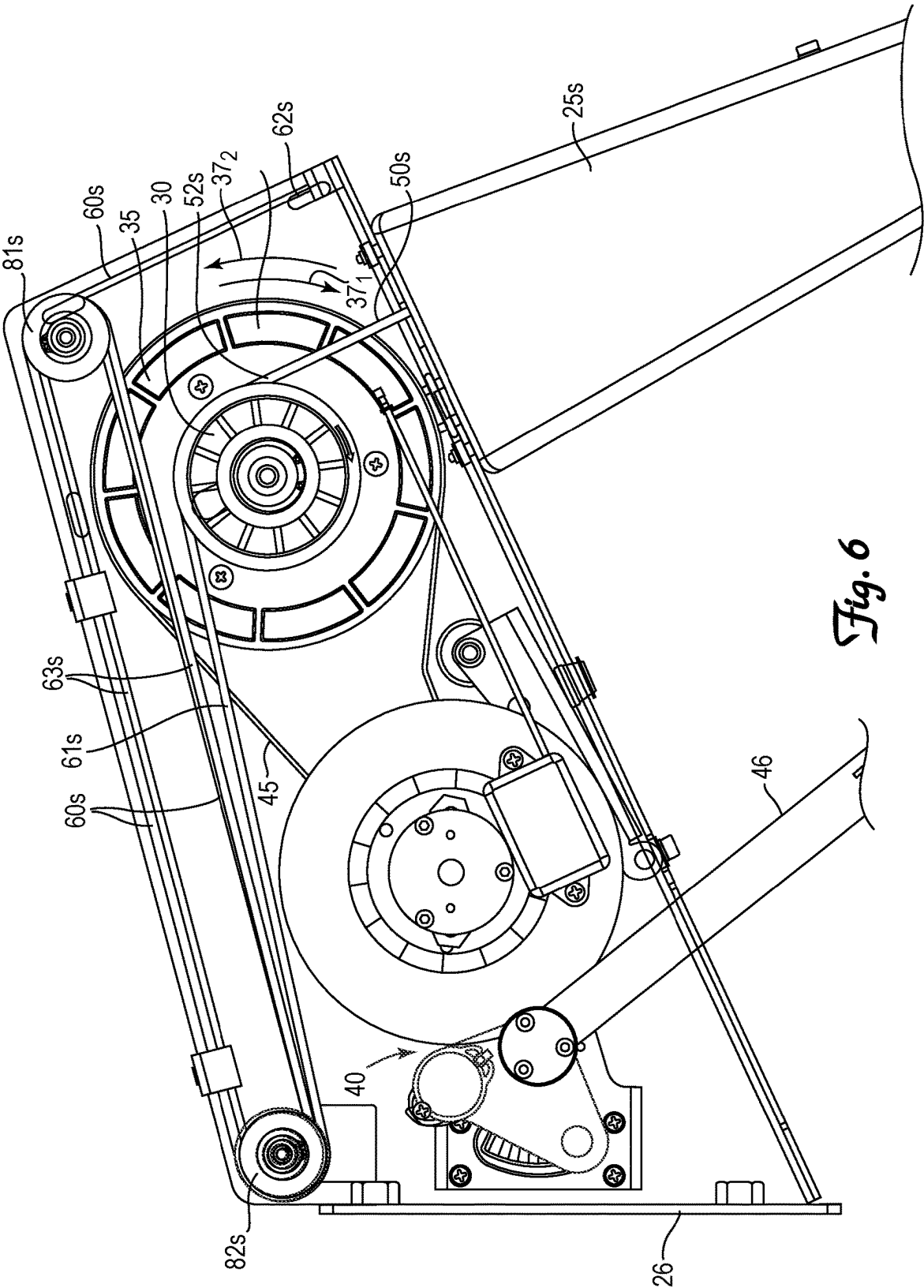


Fig. 5



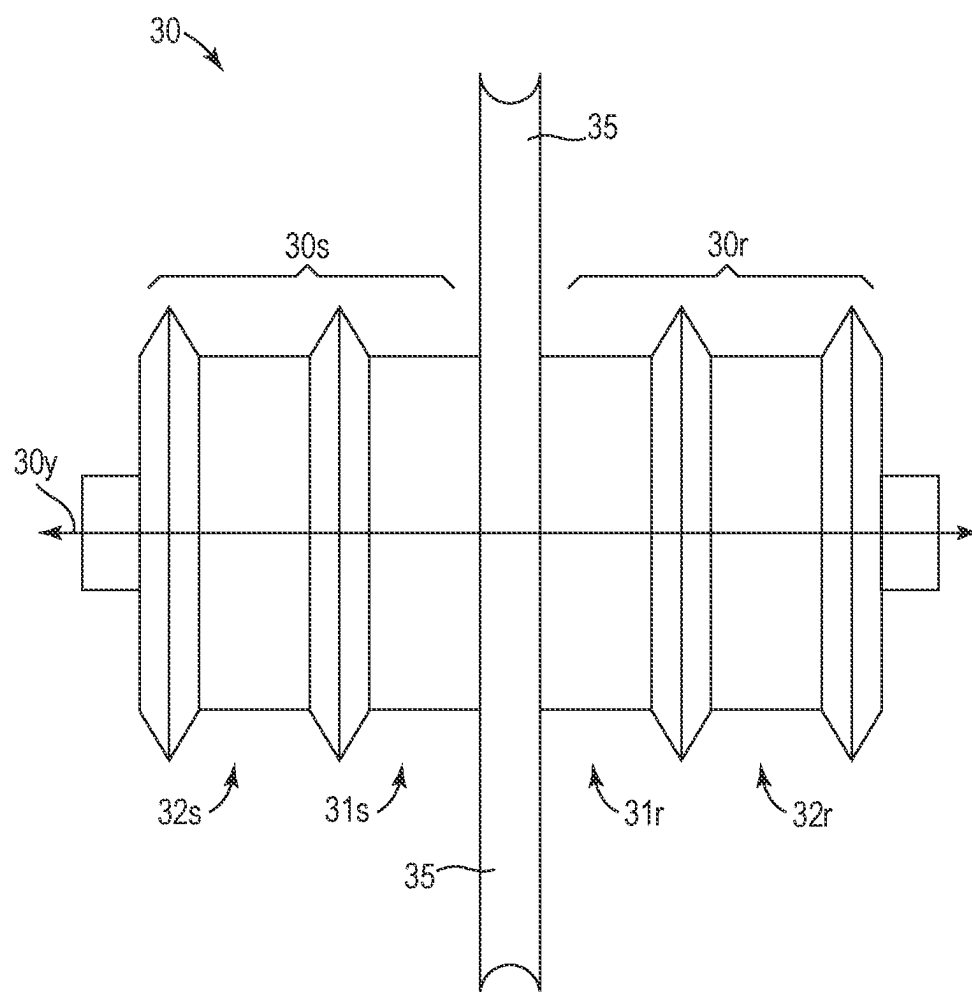


Fig. 7

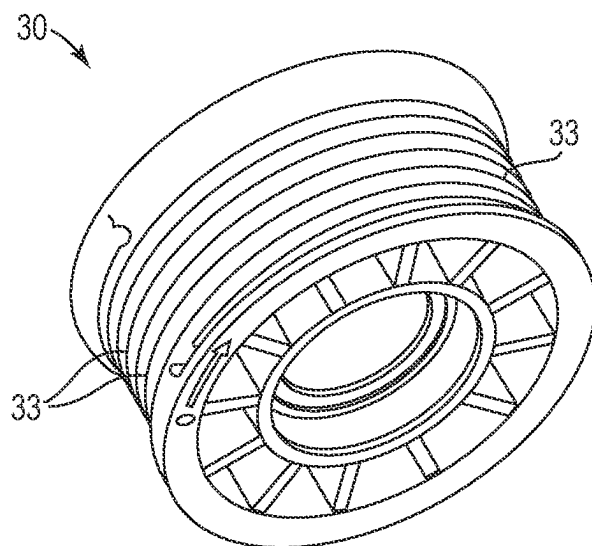


Fig. 8

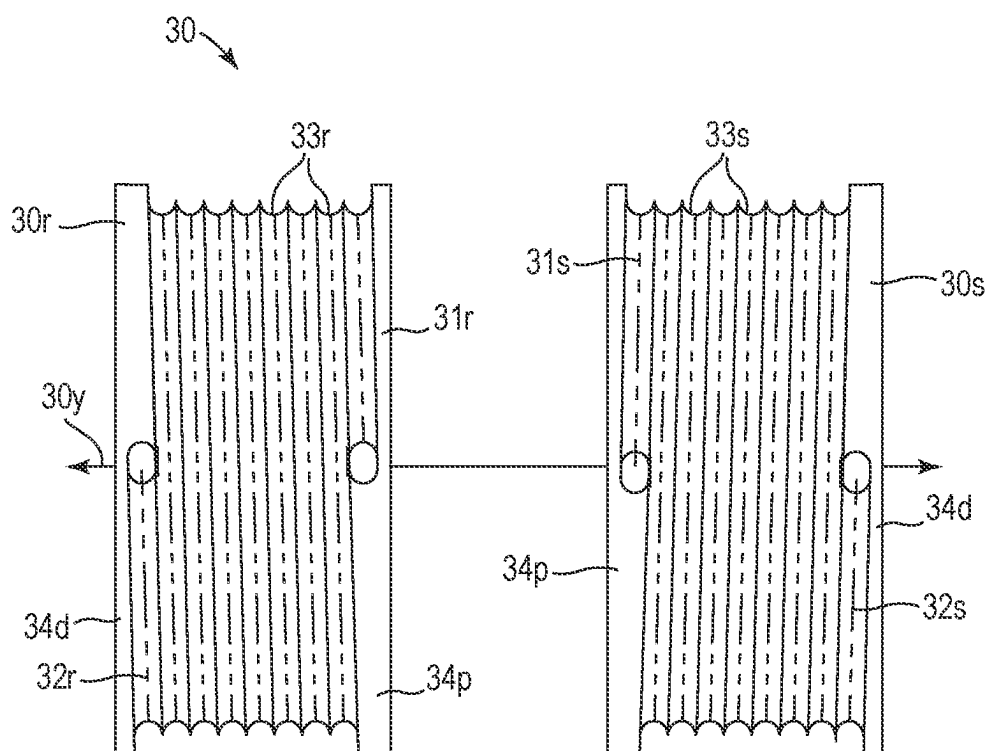


Fig. 8A

UPPER BODY SKI MOTION EXERCISE MACHINE

BACKGROUND OF THE INVENTION

[0001] Variety in exercise options is an important aspect in maintaining an individual's interest and desire to continue exercising. It is also imperative that the equipment necessary for providing each of the exercise options have a minimal footprint and require modest space to perform the exercise so that the various pieces of equipment necessary to provide a wide variety of exercise options and the corresponding space required to utilize the equipment may be provided within a reasonably sized commercial fitness club or home exercise room.

[0002] Hence, a continuing need exists for exercise machines that offer new, novel, fresh, and innovative approaches and techniques for exercising, particularly when combined with a modest footprint for the machines and modest required exercising space.

SUMMARY OF THE INVENTION

[0003] The invention is an upper body ski motion exercise machine.

[0004] A first embodiment of the invention includes (i) a pair of spools, (ii) a single means for applying resistance to rotation of the pair of spools, (iii) right and left side inelastic lines, (iv) right and left side elastic lines, and (v) a pair of grips.

[0005] The pair of spools are laterally spaced right side and left side spools, each rotatable about a respective lateral axis of rotation. The single means for applying resistance to rotation of the pair of spools provides resistance to rotation of each spool about a respective lateral axis of rotation in a rotational direction, sans application of resistance to rotation of the pair of spools about the respective lateral axis of rotation in a counter rotational direction. The right side inelastic line has a first end restrained for a spaced relationship relative to the right spool lateral axis of rotation, and a second end secured to the right side spool. The left side inelastic line has a first end restrained for a spaced relationship relative to the left spool lateral axis of rotation, and a second end secured to the left side spool. The right side elastic line has a primary end secured to the right side spool and a secondary end restrained for maintaining a spaced relationship relative to the right side spool lateral axis of rotation. The left side elastic line has a primary end secured to the left side spool and a secondary end restrained for maintaining a spaced relationship relative to the left side spool lateral axis of rotation. The pair of grips include a right grip secured to the first end of the right side inelastic line, and a left grip secured to the first end of the left side inelastic line.

[0006] The right side inelastic line and right side elastic line are each configured and arranged relative to the right side spool so that rotation and counter-rotation of the right side spool effects corresponding winding and unwinding of the right side inelastic line on the right side spool with inverse winding and unwinding of the right side elastic line on the right side spool. In similar fashion, the left side inelastic line and left side elastic line are each configured and arranged relative to the left side spool so that rotation and counter-rotation of the left side spool effects corresponding winding and unwinding of the left side inelastic line on

the left side spool with inverse winding and unwinding of the left side elastic line on the left side spool.

[0007] A second embodiment of the invention includes (i) a pair of spools, (ii) a single means for applying resistance to rotation of the pair of spools, (iii) right and left side inelastic lines, (iv) right and left side elastic lines, and (v) a pair of grips.

[0008] The pair of spools are laterally spaced right side and left side spools each rotatable about a respective lateral axis of rotation. The single means for applying resistance to rotation of the pair of spools provides resistance to rotation of each spool about a respective lateral axis of rotation in a rotational direction, sans application of resistance to rotation of the pair of spools about the respective lateral axis of rotation in a counter rotational direction.

[0009] The right side inelastic line has a first end restrained for a spaced relationship relative to the right spool lateral axis of rotation, and a second end secured to the right side spool for (i) winding a length of the right side inelastic line proximate the second end of the right side inelastic line onto the right side spool when the right side spool is rotated in the counter rotational direction, and (ii) unwinding a length of the right side inelastic line proximate the second end of the right side inelastic line from the right side spool against the resistance applied by the single means for applying resistance when the right side spool is rotated in the rotational direction by a force pulling the first end of the right side inelastic line away from the right spool lateral axis of rotation. In similar fashion, the left side inelastic line has a first end restrained for a spaced relationship relative to the left spool lateral axis of rotation and a second end secured to the left side spool for (i) winding a length of the left side inelastic line proximate the second end of the left side inelastic line onto the left side spool when the left side spool is rotated in the counter rotational direction, and (ii) unwinding a length of the left side inelastic line proximate the second end of the left side inelastic line from the left side spool against the resistance applied by the single means for applying resistance when the left side spool is rotated in the rotational direction by a force pulling the first end of the left side inelastic line away from the left spool lateral axis of rotation. The pair of grips include a right grip secured to the first end of the right side inelastic line, and a left grip secured to the first end of the left side inelastic line.

[0010] The right side elastic line has a secondary end restrained for maintaining a spaced relationship relative to the right side spool lateral axis of rotation, and a primary end secured to the right side spool whereby a length of the right side elastic line proximate the primary end of the right side elastic line is operable for (i) winding onto the right side spool when the right side spool is rotated in the rotational direction so as to stretch the right side elastic line and thereby create a biasing force favoring rotation of the right side spool in the counter rotation direction, and (ii) unwinding from the right side spool when the right side spool is counter rotated by the bias of the stretched right side elastic line. In similar fashion, a left side elastic line has a secondary end restrained for maintaining a spaced relationship relative to the left side spool lateral axis of rotation, and a primary end secured to the left side spool whereby a length of the left side elastic line proximate the primary end is operable for (i) winding onto the left side spool when the left side spool is rotated in the rotational direction so as to stretch the left side elastic line and thereby create a biasing force

favoring rotation of the left side spool in the counter rotation direction, and (ii) unwinding from the left side spool when the left side spool is counter rotated by the bias of the stretched left side elastic line.

[0011] The upper body ski motion exercise machine can include a spaced pair of right pulleys with an intermediate length of the right elastic line looped repeatedly around the pair of right pulleys, and a spaced pair of left pulleys with an intermediate length of the left elastic line looped repeatedly around the pair of left pulleys.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. 1 is a perspective view of one embodiment of the invention.

[0013] FIG. 2 is a side view of the invention depicted in FIG. 1.

[0014] FIG. 3 is a front view of the invention depicted in FIG. 1.

[0015] FIG. 4 is a perspective view of the invention depicted in FIG. 1 with the housing removed to facilitate depiction of the internal components.

[0016] FIG. 5 is a right side view of the invention depicted in FIG. 4 with the right side panel of the housing removed to facilitate depiction of the internal components.

[0017] FIG. 6 is a left side view of the invention depicted in FIG. 4 with the left side panel of the housing removed to facilitate depiction of the internal components.

[0018] FIG. 7 is a top view of one embodiment of the spool component of the invention depicted in FIG. 4.

[0019] FIG. 8 is a perspective view of one side of an alternative embodiment of the spool component of the invention depicted in FIG. 4.

[0020] FIG. 8A is a rear view of both right and left sides of the alternative embodiment of the spool component depicted in FIG. 8, with the option of the elastic line secured to the distal lateral end of each spool and the inelastic line secured to the proximal lateral end of each spool.

DETAILED DESCRIPTION OF AN ILLUSTRATIVE EMBODIMENT OF THE INVENTION

Definitions

[0021] As utilized herein, the term compact, in reference to the exercise machine in accordance with this invention, means an exercise machine that when in a rest or inactive mode has all moving components (i.e., components that translate, expand, contract, rotate and/or pivot during use of the machine excluding components that are optionally moved when customizing machine settings such as resistance level or location of the display) that can either (i) fit within a single rectangular cuboid having a longitudinal depth of less than 3 feet, a lateral width of less than 3 feet and a transverse vertical height of less than 4 feet or (ii) fit within a single hexahedron having a volume less than 30 ft³. In similar fashion, as utilized herein, the term ultra compact an exercise machine that when in a rest or inactive mode has all moving components (i.e., components that translate, expand, contract, rotate and/or pivot during use of the machine excluding components that are optionally moved when customizing machine settings such as resistance level or location of the display) that can either (i) fitting within a single rectangular cuboid having a longitudinal depth of less

than 2.5 feet, a lateral width of less than 2.5 feet and a transverse vertical height of less than 3 feet, or (ii) fitting within a single hexahedron having a volume less than 20 ft³.

[0022] As utilized herein, the term elevated, in reference to an exercise machine in accordance with this invention, means an exercise machine that when stably supported upon a floor in a position and configuration for immediate use, and when in a rest or inactive mode, has all moving components (i.e., components that translate, expand, contract, rotate and/or pivot during use of the machine excluding components that are optionally moved when customizing machine settings such as resistance level or location of the display) at least 4 feet above the floor. In similar fashion, as utilized herein, the term peak elevated means all moving components are at least 5 feet above the floor.

[0023] As utilized herein, the phrase top mount, in reference to an exercise machine in accordance with this invention, means an exercise machine that when in a rest or inactive mode has the grips positioned vertically below all other moving components (i.e., components that translate, expand, contract, rotate and/or pivot during use of the machine excluding components that are optionally moved when customizing machine settings such as resistance level or location of the display).

[0024] As utilized herein, the directional terms right and left are made from the perspective of a user facing the front of the exercise machine.

Nomenclature Table

REF. NO.	NAME
10	Upper Body Ski Motion Exercise Machine
20	Housing
20r	Right Side of Housing
20s	Left Side of Housing
20x ₁	Front of Housing
20x ₂	Back of Housing
20z ₁	Top of Housing
20z ₂	Bottom of Housing
25	Guide Frame
25r	Right Side Arm of Guide Frame
25s	Left Side Arm of Guide Frame
25t	Cross Member of Guide Frame
26	Mounting Bracket
30	Spools
30r	Right Side Spool
30s	Left Side Spool
30y	Spool Axis of Rotation
31r	Right Side Spool Inelastic Line Retention Zone
32r	Right Side Spool Elastic Line Retention Zone
31s	Left Side Spool Inelastic Line Retention Zone
32s	Left Side Spool Elastic Line Retention Zone
33	Helical Groove on the Spools
33r	Right Side Spool Shared Helical Groove
33s	Left Side Spool Shared Helical Groove
34d	Distal Lateral End of Each of Right and Left Side Spools
34p	Proximal Lateral End of Each of Right and Left Side Spools
35	Resistance Communicating Pulley
37 ₁	Rotational Direction Around Spool Axis of Rotation
37 ₂	Counter Rotational Direction Around Spool Axis of Rotation
40	Resistance or Braking Device
45	Belt Transmitting Resistance from Resistance Device to Spools
46	Resistance Level Adjustment Lever
50	Inelastic Lines 50r and 50s
50r	Right Side Inelastic Line
51r	First End of Right Side Inelastic Line
52r	Second End of Right Side Inelastic Line
50s	Left Side Inelastic Line
51s	First End of Left Side Inelastic Line

-continued

Nomenclature Table

REF. NO.	NAME
52s	Second End of Left Side Inelastic Line
60	Elastic Lines 60r and 60s
60r	Right Side Elastic Line
61r	Primary End of Right Side Elastic Line
62r	Secondary End of Right Side Elastic Line
63r	Intermediate Length of Right Side Elastic Line
60s	Left Side Elastic Line
61s	Primary End of Left Side Elastic Line
62s	Secondary End of Left Side Elastic Line
63s	Intermediate Length of Left Side Elastic Line
70	Grips 70r and 70s
70r	Right Grip
70s	Left Grip
80	Looping Pulleys 81 and 82
81	First Looping Pulleys
81r	Right Side First Looping Pulley
81s	Left Side First Looping Pulley
82	Second Looping Pulleys
82r	Right Side Second Looping Pulley
82s	Left Side Second Looping Pulley
90	Guide Pulleys 91 and 92
91	First Guide Pulleys 91r and 91s
91r	First Right Side Guide Pulley
91rz	First Right Side Guide Pulley Transverse Axis of Rotation
91s	First Left Side Guide Pulley
91sz	First Left Side Guide Pulley Transverse Axis of Rotation
92	Second Guide Pulleys 92r and 92s
92r	Second Right Side Guide Pulley
92rz	Second Right Side Guide Pulley Transverse Axis of Rotation
92s	Second Left Side Guide Pulley
92sz	Second Left Side Guide Pulley Transverse Axis of Rotation
93	Second Guide Pulley Mounting Fixtures 93r and 93s
93r	Right Side Second Guide Pulley Mounting Fixture
93s	Left Side Second Guide Pulley Mounting Fixture
94	Hinges on Second Guide Pulley Mounting Fixtures 94r and 94s
94r	Hinge on Right Side Second Guide Pulley Mounting Fixture
94ry	Lateral Pivot Axis of Right Side Hinge
94s	Hinge on Left Side Second Guide Pulley Mounting Fixture
94sy	Lateral Pivot Axis of Left Side Hinge
100	Articulating Linkage
101	First End of Articulating Linkage
102	Second End of Articulating Linkage
120	Display
x	Longitudinal (Front to Back or Depth) Direction and Dimension
y	Lateral (Side to Side or Width) Direction and Dimension
z	Transverse (Top to Bottom or Height) Direction and Dimension

Construction

[0025] Referring to FIGS. 1-6, the invention is an upper body ski motion exercise machine 10 and most preferably a compact upper body ski motion exercise machine 10. A particular embodiment of the exercise machine 10 includes a frame or preferably an enclosing housing 20, a pair of spools 30, a resistance device 40, right side and left side inelastic lines 50r and 50s, right side and left side elastic lines 60r and 60s, a pair of grips 70, a pair of looping pulleys 80, and guide pulleys 90 such as a pair of first guide pulleys 91, and a pair of second guide pulleys 92.

[0026] Referring to FIGS. 1-3, the housing 20 includes a right side panel 20r, a left side panel 20s, a front panel 20x₁, a rear panel 20x₂, a top panel 20z₁ and a bottom panel 20z₂. The housing 20 can protectively and aesthetically encompass and shroud the pair of spools 30, the resistance device 40, the right side and left side inelastic lines 50r and 50s, the

right side and left side elastic lines 60r and 60s, and the pair of looping pulleys 80. The housing 20 also provides anchoring points for these various components.

[0027] Referring to FIGS. 1-4, a guide frame 25 can extend longitudinally x down from the bottom 20z₂ of the housing 20, having laterally y diverging right side and left side guide arms 25r and 25s for supporting a right and left lateral y spacing of right and left grips 70r and 70s. The lateral y spacing between the right and left grips 70r and 70s should allow a user's arms to extend parallel to one another when holding the grips 70 proximate the guide arms 25r and 25s. A lateral y spacing of about 1.5 to 2.5 feet should generally be effective. A lateral y cross member 25t can connect the free terminal end portions (unnumbered) of the right and left guide arms 25r and 25s to improve the structural integrity of the guide frame 25.

[0028] Referring to FIGS. 1 and 4, a mounting bracket 26 for mounting the upper body ski motion exercise machine 10 to a vertical structure such as a wall or an exercise frame (not shown) can be provided at the back 20x₂ of the housing 20. When mounted, the upper body ski motion exercise machine 10 is preferably an elevated exercise machine and more preferably a top mount elevated exercise machine.

[0029] Referring to FIG. 7, the pair of spools 30 include laterally y spaced right side and left side spools 30r and 30s, each rotatably secured to the housing 20 and rotatable in rotational 37₁ and counter-rotational 37₂ directions about a respective lateral y axis of rotation 30y. The axis of rotation 30y may be different for each of the right side and left side spools 30r and 30s, but as depicted in FIG. 7, the right side and left side spools 30r and 30s preferably rotate about a common axis of rotation 30y.

[0030] Still referring to FIG. 7, each of the right side and left side spools 30r and 30s can be divided into two distinct laterally y separated retention zones for providing separate volumes for retention of inelastic 50 and elastic 60 lines wound onto the spools 30. The right side spool 30r can be divided by a circumferential shoulder (unnumbered) into a right side spool 30r inelastic line retention zone 31r and a right side spool 30r elastic line retention zone 32r. In mirror image fashion, the left side spool 30s can be divided by a circumferential shoulder (unnumbered) into a left side spool 30s inelastic line retention zone 31s and a left side spool 30s elastic line retention zone 32s.

[0031] Referring to FIGS. 8 and 8A, an alternative embodiment of the spools 30 for guided winding and unwinding of the inelastic 50 and elastic 60 lines onto and off from the spools 30 employs right side and left side spools 30r and 30s each of which have a single continuous helical groove 33r and 33s respectively, shared by the corresponding set of right side inelastic 50r and elastic 60r lines on the right side spool 30r and corresponding set of left side inelastic 50s and left side elastic 60s lines on the left side spool 30r. One of the lines 50 or 60 on each of the spools 30 has an end (i.e., the second end 52 of the inelastic line 50 or the primary end 61 of the elastic line 60) secured immediate the proximal lateral end 34p of the helical groove 33 on each spool 30 which is wound in a clockwise direction onto the helical groove 33 while the other of the lines 50 or 60 on the spools 30 has an end (i.e., the second end 52 of the inelastic line 50 or the primary end 61 of the elastic line 60) secured immediate the distal lateral end 34d of the helical groove 33 on each spool 30 which is wound in a counter-clockwise direction onto the helical groove 33. This configuration of

laterally y spaced connection of the inelastic **50** and elastic **60** lines to each spool **30** and counter windings of the inelastic **50** and elastic **60** lines onto each spool **30** allows the lines **50** and **60** on each spool **30** to share the same helical groove **33** because as one of the lines **50** or **60** winds onto the helical groove **33** the other line **50** or **60** unwinds from the helical groove **33**, leaving windings of the groove **33** open and available for occupancy by the line **50** or **60** winding onto the helical groove **33**. Sharing of a single continuous helical groove **33r** and **33s** on each spool **30r** and **30s** minimizes the overall lateral y width of the exercise machine **10** and adds to the compact nature of the machine **10**.

[0032] Referring to FIGS. 4-6, the means for applying resistance **40** to rotation of the pair of spools **30** about the respective lateral y axis of rotation **30y** is rotatably secured to the housing **20** and provides resistance in only one rotational direction (e.g., a clutched brake for applying resistance in the rotational direction **37₁** but not in the counter-rotational direction **37₂**). The means for applying resistance **40** transmits the resistance to the pair of spools **30** via a belt **45** that loops around a pulley (unnumbered) on the means for applying resistance **40** and around a resistance communicating pulley **35** concentrically affixed to the pair of spools **30**. The means for applying resistance **40** can be selected from any of the various braking devices used upon exercise machines, including, specifically but not exclusively, friction, hydraulic, magnetic, electromagnetic, air resistance, etc.

[0033] Referring to FIGS. 1-6, a lever **46** or equivalent mechanism, such as a pull cord or dial, may be provided to allow user adjustment to the resistance level. Referring to FIGS. 1 and 2, an indicia plate (unnumbered) with resistance level markings viewable from the front of the exercise machine **10** can be provided.

[0034] Referring to FIGS. 1, 3 and 4, the right side and left side inelastic lines **50r** and **50s** are mirror images of one another. As the name implies, both are inelastic lengths of rope, cable, cord or similar material.

[0035] The right side inelastic line **50r** has a first end **51r** and a second end **52r**. The first end **51r** projects out from the lower distal end (unnumbered) of the guide frame **25** proximate the right side arm **25r** with a right grip **70r** secured to the first end **51r**. The second end **52r** is secured to the right side spool **30r** for winding and unwinding of the right side inelastic line **50r** onto and from around the right side spool **30r**. To provide a desired positioning and orientation of the right grip **70r**, the right side inelastic line **50r** can be sequentially redirected from the second end **52r** by a first right side guide pulley **91r** rotatably mounted to the cross member **25t** of the guide frame **25** proximate the lateral y center of the guide frame **25** for rotation about a first right side guide pulley transverse axis of rotation **91rz** and then by a second right side guide pulley **92r** rotatably mounted to the guide frame **25** proximate the lower distal end of the guide frame **25** proximate the right side arm **25r** for rotation about a second right side guide pulley transverse axis of rotation **92rz**.

[0036] The first end **51r** can be conveniently restrained from being pulled into contact with the second right side guide pulley **92r** by configuring a right side second guide pulley mounting fixture **93r** supporting and encompassing the second right side guide pulley **92r** so as to allow passage

of the right side inelastic line **50r** into engagement with the second right side guide pulley **92r** but not the right grip **70r**.

[0037] In mirror fashion, the left side inelastic line **50s** has a first end **51s** and a second end **52s**. The first end **51s** projects out from the lower distal end (unnumbered) of the guide frame **25** proximate the left side arm **25s** with a left grip **70s** secured to the first end **51s**. The second end **52s** is secured to the left side spool **30s** for winding and unwinding of the left side inelastic line **50s** onto and from around the left side spool **30s**. To provide a desired positioning and orientation of the left grip **70s**, the left side inelastic line **50s** can be sequentially redirected from the second end **52s** by a first left side guide pulley **91s** rotatably mounted to the cross member **25t** of the guide frame **25** proximate the lateral y center of the guide frame **25** for rotation about a first left side guide pulley transverse z axis of rotation **91sz** and then by a second left side guide pulley **92s** rotatably mounted to the guide frame **25** proximate the lower distal end of the guide frame **25** proximate the left side arm **25s** for rotation about a second left side guide pulley transverse axis of rotation **92sz**.

[0038] The first end **51s** can be conveniently restrained from being pulled into contact with the second left side guide pulley **92s** by configuring a left side second guide pulley mounting fixture **93s** supporting and encompassing the second left side guide pulley **92s** so as to allow passage of the left side inelastic line **50s** into engagement with the second left side guide pulley **92s** but not the left grip **70s**.

[0039] Referring to FIGS. 1-4, each of the right side and left side second guide pulley mounting fixtures **93r** and **93s** can be attached to the guide frame **25** by hinges **94r** and **94s** respectively, with each hinge **94r** and **94s** allowing pivoting of the associated second guide pulley mounting fixtures **93r** and **93s** and thereby the associated second guide pulleys **92r** and **92s** about a lateral y pivot axis **94ry** and **94sy** respectively. This accommodates pulling of the grips **70** in the longitudinal x direction away from the guide frame **25** at different angles relative to the transverse direction z.

[0040] Referring to FIGS. 4-6, the right side and left side elastic lines **60r** and **60s** are mirror images of one another. As the name implies, both are elastic lengths of rope, cord or similar material, commonly generically referenced as bungee cords.

[0041] To provide a length of each of the right and left side elastic lines **60r** and **60s** capable of accommodating a full range of motion to the right and left grips **70r** and **70s** respectively, as necessary to allow emulation of upper body and right arm movement during skiing from a compact housing **20**, an intermediate length **63r** and **63s** of each of the right side and left side elastic lines **60r** and **60s** respectively, may be looped one or more times around spaced first and second looping pulleys **81** and **82**.

[0042] The right side elastic line **60r** has a primary end **61r** and a secondary end **62r**. The secondary end **62r** is secured to the housing **20**. The primary end **61r** is secured to the right side spool **30r** for winding and unwinding of the right side elastic line **60r** onto and from around the right side spool **30r**. To provide a length of the right side elastic line **60r** capable of accommodating a full range of motion to the right grip **70r** as necessary to allow emulation of upper body and right arm movement during skiing from a compact housing **20**, an intermediate length **63r** of the right side elastic line **60r** may be looped one or more times around and between a first pair

of spaced right side first and second looping pulleys **81r** and **82r** with a preference for two to five loops.

[0043] In mirror fashion, the left side elastic line **60s** has a primary end **61s** and a secondary end **62s**. The secondary end **62s** is secured to the housing **20**. The primary end **61s** is secured to the left side spool **30s** for winding and unwinding of the left side elastic line **60s** onto and from around the left side spool **30s**. To provide a length of the left side elastic line **60s** capable of accommodating a full range of motion to the left grip **70s** as necessary to allow emulation of upper body and left arm movement during skiing from a compact housing **20**, an intermediate length **63s** of the left side elastic line **60s** may be looped one or more times around and between a pair of spaced left side first and second looping pulleys **81s** and **82s** with a preference for two to five loops.

[0044] The inelastic lines **50** and elastic lines **60** are secured to the spools **30** so that as one is unwound from the spools **30** the other is wound onto the spools **30** and vice versa. More specifically, pulling downward on the right and left grips **70r** and **70s** with sufficient force to overcome the resistance exerted by the resistance device **40** and translate the grips **70** away from the spools **30** causes the right and left inelastic lines **50r** and **50s** to unwind from the right and left spools **30r** and **30s** respectively, rotating the spools **30r** and **30s** and causing the right and left elastic lines **60r** and **60s** to wind onto the right and left spools **30r** and **30s** respectively. This winding of the right and left elastic lines **60r** and **60s** onto the right and left spools **30r** and **30s** respectively creates a biasing counter-rotation **37₂** pull force in the elastic lines **60**. Upon diminishment of the pulling force below the biasing pull force in the elastic lines **60r** and **60s**, the biasing pull force can effect counter-rotation **37₂** of the spools **30r** and **30s** and thereby unwind the right and left elastic lines **60r** and **60s** from the right and left spools **30r** and **30s** with concomitant diminishment of the biasing force, and with concomitant winding of the right and left inelastic lines **50r** and **50s** onto the right and left spools **30r** and **30s** respectively, until further return of the right and left inelastic lines **50r** and **50s** is prevented such as by the right and left grips **70r** and **70s** engaging the second guide pulley mounting fixtures **93r** and **93s** respectively.

[0045] Referring to FIGS. 1-3, a display **120** in communication with sensors (not shown) on the machine **10** can be provided for displaying relevant selected exercise parameters and performance details to a user during exercise, such as distance, power, calories burned, elapsed time, etc. One suitable embodiment is attaching the display **120** to the exercise machine **10** via a repositionable articulating linkage **100** attached at a first end **101** to the guide frame **25** with the display **120** attached to the second end **102**.

We claim:

1. An upper body ski motion exercise machine, comprising:

- (a) a pair of laterally spaced right side and left side spools each rotatable about a respective lateral axis of rotation,
- (b) a single means for applying resistance to rotation of the pair of spools about the respective lateral axis of rotation in a rotational direction, sans application of resistance to rotation of the pair of spools about the respective lateral axis of rotation in a counter rotational direction,
- (c) a right side inelastic line having a first end and a second end wherein the first end of the right side

inelastic line is restrained for a spaced relationship relative to the right spool lateral axis of rotation and the second end of the right side inelastic line is secured to the right side spool,

- (d) a left side inelastic line having a first end and a second end wherein the first end of the left side inelastic line is restrained for a spaced relationship relative to the left spool lateral axis of rotation and the second end of the left side inelastic line is secured to the left side spool,
- (e) a right grip secured to the first end of the right side inelastic line,
- (f) a left grip secured to the first end of the left side inelastic line,
- (g) a right side elastic line having a primary end and a secondary end wherein the secondary end of the right side elastic line is restrained for maintaining a spaced relationship relative to the right side spool lateral axis of rotation, and the primary end of the right side elastic line is secured to the right side spool,
- (h) a left side elastic line having a primary end and a secondary end wherein the secondary end of the left side elastic line is restrained for maintaining a spaced relationship relative to the left side spool lateral axis of rotation, and the primary end of the left side elastic line is secured to the left side spool,
- (i) wherein the right side inelastic line and right side elastic line are each configured and arranged relative to the right side spool that rotation and counter-rotation of the right side spool effects corresponding winding and unwinding of the right side inelastic line on the right side spool with inverse winding and unwinding of the right side elastic line on the right side spool, and
- (j) wherein the left side inelastic line and left side elastic line are each configured and arranged relative to the left side spool that rotation and counter-rotation of the left side spool effects corresponding winding and unwinding of the left side inelastic line on the left side spool with inverse winding and unwinding of the left side elastic line on the left side spool.

2. The upper body ski motion exercise machine of claim 1 wherein the exercise machine is compact.

3. The upper body ski motion exercise machine of claim 1 wherein the exercise machine is elevated.

4. The upper body ski motion exercise machine of claim 1 wherein the exercise machine is a top mount exercise machine.

5. The upper body ski motion exercise machine of claim 1 wherein the pair of spools are rotatable in tandem about a common lateral axis of rotation.

6. An upper body ski motion exercise machine, comprising:

- (a) a pair of laterally spaced right side and left side spools each rotatable about a respective lateral axis of rotation,
- (b) a single means for applying resistance to rotation of the pair of spools about the respective lateral axis of rotation in a rotational direction, sans application of resistance to rotation of the pair of spools about the respective lateral axis of rotation in a counter rotational direction,
- (c) a right side inelastic line having a first end and a second end wherein the first end of the right side inelastic line is restrained for a spaced relationship relative to the right spool lateral axis of rotation and the second end of the right side inelastic line is secured to

- the right side spool for (i) winding a length of the right side inelastic line proximate the second end of the right side inelastic line onto the right side spool when the right side spool is rotated in the counter rotational direction, and (ii) unwinding a length of the right side inelastic line proximate the second end of the right side inelastic line from the right side spool against the resistance applied by the at least one means for applying resistance when the right side spool is rotated in the rotational direction by a force pulling the first end of the right side inelastic line away from the right spool lateral axis of rotation,
- (d) a left side inelastic line having a first end and a second end wherein the first end of the left side inelastic line is restrained for a spaced relationship relative to the left spool lateral axis of rotation and the second end of the left side inelastic line is secured to the left side spool for (i) winding a length of the left side inelastic line proximate the second end of the left side inelastic line onto the left side spool when the left side spool is rotated in the counter rotational direction, and (ii) unwinding a length of the left side inelastic line proximate the second end of the left side inelastic line from the left side spool against the resistance applied by the at least one means for applying resistance when the left side spool is rotated in the rotational direction by a force pulling the first end of the left side inelastic line away from the left spool lateral axis of rotation,
- (e) a right grip secured to the first end of the right side inelastic line,
- (f) a left grip secured to the first end of the left side inelastic line,
- (g) a right side elastic line having a primary end and a secondary end wherein the secondary end of the right side elastic line is restrained for maintaining a spaced relationship relative to the right side spool lateral axis of rotation, and the primary end of the right side elastic line is secured to the right side spool whereby a length of the right side elastic line proximate the primary end of the right side elastic line is operable for (i) winding onto the right side spool when the right side spool is rotated in the rotational direction so as to stretch the right side elastic line and thereby create a biasing force favoring rotation of the right side spool in the counter rotation direction, and (ii) unwinding from the right side spool when the right side spool is counter rotated by the bias of the stretched right side elastic line,
- (h) a left side elastic line having a primary end and a secondary end wherein the secondary end of the left side elastic line is restrained for maintaining a spaced relationship relative to the left side spool lateral axis of rotation, and the primary end of the left side elastic line is secured to the left side spool whereby a length of the left side elastic line proximate the primary end of the left side elastic line is operable for (i) winding onto the left side spool when the left side spool is rotated in the rotational direction so as to stretch the left side elastic line and thereby create a biasing force favoring rotation of the left side spool in the counter rotation direction, and (ii) unwinding from the left side spool when the left side spool is counter rotated by the bias of the stretched left side elastic line.
7. The upper body ski motion exercise machine of claim 6 wherein the exercise machine is compact.
8. The upper body ski motion exercise machine of claim 6 wherein the exercise machine is elevated.
9. The upper body ski motion exercise machine of claim 6 wherein the exercise machine is a top mount exercise machine.
10. The upper body ski motion exercise machine of claim 6 wherein the pair of spools are rotatable in tandem about a common lateral axis of rotation.
11. The upper body ski motion exercise machine of claim 6 wherein the right side spool and the left side spool rotate independently from one another in both the rotational and counter rotational directions.
12. The upper body ski motion exercise machine of claim 6 wherein each of the right and left elastic lines have an intermediate length between the primary and secondary ends looped repeatedly around and between an associated pair of right and left spaced pulleys respectively.
13. The upper body ski motion exercise machine of claim 12 wherein the intermediate length of each of the right side and left side elastic lines are looped at least three times around and between the associated pair of right and left spaced pulleys respectively.
14. The upper body ski motion exercise machine of claim 6 wherein:
- (a) the right side inelastic line has a length from the first end to the second end, the right side elastic line has a length from the primary end to the secondary end, and the length of the right side elastic line is at least twice the length of the right side inelastic line, and
- (b) the left side inelastic line has a length from the first end to the second end, the left side elastic line has a length from the primary end to the secondary end, and the length of the left side elastic line is at least twice the length of the left side inelastic line.
15. The upper body ski motion exercise machine of claim 6 wherein:
- (a) the right side spool is laterally divided into separate first and second right side zones for individually and separately retaining wound right side inelastic line in the first right side zone and right side elastic line in the second right side zone, and
- (b) the left side spool is laterally divided into separate first and second left side zones for individually and separately retaining wound left side inelastic line in the first left side zone and left side elastic line in the second left side zone.
16. The upper body ski motion exercise machine of claim 6 wherein the right side and left side spools each have a single continuous helical groove shared by the corresponding set of inelastic and elastic lines with inverse winding and unwinding of the inelastic and elastic lines on each spool.
17. The upper body ski motion exercise machine of claim 6 further comprising a mounting bracket for mounting the upper body ski motion exercise machine to a vertical structure in an elevated position with the right and left grips positioned vertically below the right side and left side spools and the resistance means.